

Report on the outcomes of a Short-Term Scientific Mission¹

Action number: CA21119

Grantee name: Enes Abanoz

Details of the STSM

Title:

Silicon Sampling: Evaluating LLM Mimicry of Survey Demographics for Computational Opinion Research

Description of the work carried out during the STSM

During the STSM, the primary focus was establishing the data foundation, preprocessing pipelines, and experimental design for the "Silicon Sampling" framework.

A key deviation from the initial plan was the transition from using Eurobarometer data to using the European Social Survey (ESS) data. We decided to utilize the ESS11 dataset (integrated file, edition 4.1) because of its extensive cross-national scope, academic rigor, and thematic relevance.

To ensure the research remained focused and robust, we selected three primary themes from the survey: "Media and Social Trust", "Politics", and "Human Values". To inform and construct the LLM personas, we extracted key variables covering respondents' backgrounds, including nationality, gender, age, language, income, marital status, and voting history.

The dataset was cleaned and preprocessed by removing rows with missing values. Additionally, to preserve statistical robustness and comparability, we filtered the data to keep countries that featured an equal or similar number of participants.

On the technical side, we worked on establishing the data pipeline and preparing the system for prompt execution. While the original plan anticipated using online LLM APIs, we decided to structure the next phases of the experiment around offline LLMs to ensure greater reproducibility, data control, and cost-effectiveness. The STSM concluded with the dataset fully prepared and ready for a pilot run.

¹ This report is submitted by the grantee to the Action MC for approval and for claiming payment of the awarded grant. The Grant Awarding Coordinator coordinates the evaluation of this report on behalf of the Action MC and instructs the Grant Holder Manager to verify the grant documents compliance and proceed with the payment.

Description of the STSM main achievements and planned follow-up activities

The STSM achieved its foundational goals of conceptualizing, structuring, and preparing the "Silicon Sampling" methodology.

Key Achievements:

1. **Data Infrastructure:** Successfully processed and balanced a high-quality subset of the ESS11 dataset, mapping complex socio-demographic profiles to target opinion themes.
2. **Methodological Design:** Formulated a structured prompting approach that integrates multi-attribute human personas (including voting, language, and income variables) to assess the depth of LLM demographic mimicry.
3. **Action Alignment:** This work directly contributes to WG2 (Tools) and WG3 (Data) by designing a potential "silicon-based" standard for opinion generation. It also addresses the WG1 (Theory) challenge by establishing a framework to test whether LLMs can mimic actual human stance and argument patterns rather than just surface-level text.

Planned Follow-Up Activities:

Moving forward, the collaboration will proceed with the following steps:

- **Pilot Phase:** Run a small-scale pilot using the cleaned ESS11 data to test prompt efficacy and refine the generation pipeline.
- **Full Experimentation:** Execute the entire simulation across at least two different offline LLM models to record and compile synthetic opinionated texts.
- **Comparative Analysis:** Compare the LLM-generated outputs against the original human responses in the ESS11 dataset, analyzing the accuracy of expressed stances and argument quality.
- **Dissemination:** Integrate the resulting protocol into the Computational Communication Lab (CCL) framework (D3.3) and draft a joint methodological paper evaluating the limits and potential of LLMs as survey proxies.